ECE551: Homework 4

Due: Monday, November 2nd @ 11:55PM

For this homework, you *must*:

- Work individually.
- Use descriptive signal names and comment your code
- Remember what you learned from the Cummings SNUG paper

1 (10pts) Synopsys Design Vision Tutorial

You are required to complete the Synopsys Tutorial listed under "Tools & Tutorials" section of the Canvas page. Once you've completed it, sign the following:

I, _____, completed the Synopsys Design Vision tutorial. If I had any problems with it, I discussed them with the TA or instructor, either in person, or through email.

2 (5pts) Project Team Formation

For the course project, you will be in a team of 4. You are encouraged to find your own team members. Once you have formed a team, **come up with a team name**, and **one person** from the group should fill out this form: https://forms.gle/LpNWmBEvqGe2JUc17

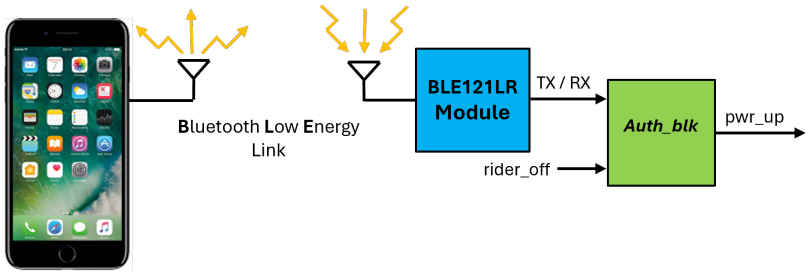
If you are **not** able to find a team, or only have a **partial team**, one person can fill out the same form **by Wednesday, October 28th**, and we will create a group for you.

Sign the following:

I, _____, have formed a 4 person project team and confirmed that one person from the group has filled out the above form. Our team name is _____.

3 (15pts) Authentication block

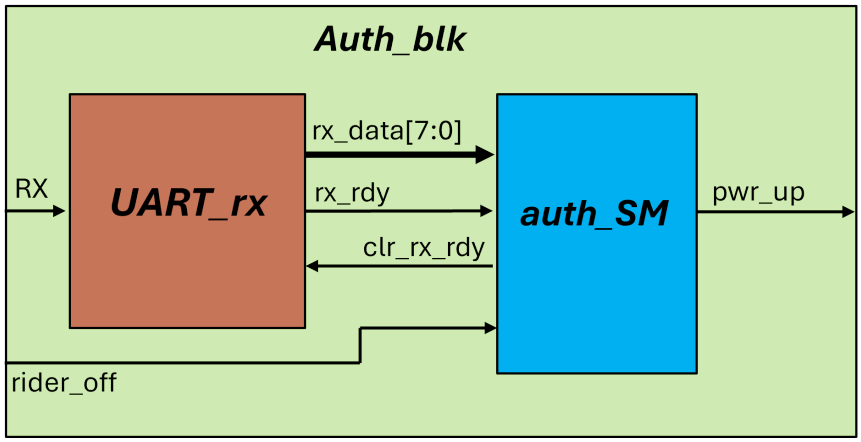
To ensure the safety of the rider, only an "authorized" rider can power on the Segway. An authorized rider will carry a phone with an app that sends the proper authorization code to a BLE module on the Segway device. The Segway control board has a BLE121LR module on that will be advertising a "Segway" service. When the user's phone scans the service, connects, and sends the appropriate authorization code it will cause the BLE121LR module to send out 0x47 ('G') over its UART TX line. This will in turn cause the **pwr_up** signal to be asserted.



When the phone app deliberately disconnects, or is disconnected due to range, the BLE121LR module sends out 0x53 ('S') over its UART TX line. The Segway then shuts down (**pwr_up**=0) if the weight on the platform no longer exceeds MIN_RIDER_WEIGHT (**rider_off** signal from en_steer block).

Of course once the "Segway" is enabled (**pwr_up** asserted) it will stay enabled as long as there is a rider on the device. We wouldn't want it to power down and throw the rider just because the phone went dead or the BLE link was interrupted.

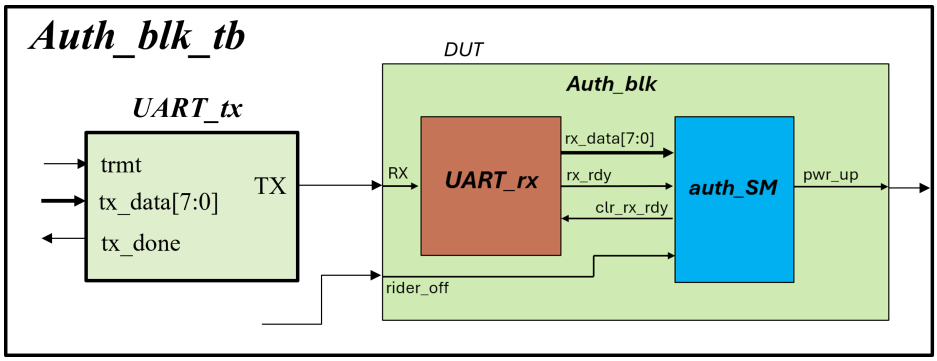
You are to design this Authentication Block (**Auth_blk**). It should include an UART receiver (**UART_rx**) and a simple state machine comparing the receptions to 'G', 'S', and monitoring the **rider_off** signal.



Draw a bubble diagram. Simulate its operation (in your head) under different scenarios.

- **pwr_up** is asserted upon reception of 'G' (0x47). It is deasserted after the last reception was 'S' and the **rider_off** signal is high.
- **UART_rx** was developed as part of HW3

Once you have developed **Auth_blk**, write a self-checking testbench for it and call it **Auth_blk_tb**.



- Remember you have a **UART_tx** from HW3 that you can use.
- Test more than the simple case of **pwr_up** and then down. Try to traverse all states and arcs of your FSM.

Submit: Auth_blk.sv, **Auth_blk_tb.sv**, and **proof** that the testbench ran and passed.

4 (20pts) Balance Control & Testing

See the Exercise 14 and 16 specification. Did you complete it well? If not, polish it up.

Submit:

- **balance_cntrl.sv**
- **PID.sv**
- **balance_cntrl_chk_tb.sv** (self-checking random vector testbench)
- **Proof** balance_cntrl.sv passed the above testbench

5 (15pts) Synthesize your balance_cntrl.sv

See Exercise 17 specification. Did you complete it well? If not, polish it up.

Submit:

- Your synthesis script (**balance_cntrl.dc**)
- The output reports for area (**area.txt**)
- The gate level verilog netlist (**balance_cntrl.vg**)

6 (35pts) SPI Transceiver

See Exercise 15 specification. Did you complete it well? If not, polish it up.

Submit:

- **SPI_main.sv**
- **SPI_main_tb.sv**
- **Proof** your DUT passed the self-checking testbench